

Unit 2: User interaction design

User-Centered Design (UCD) is the overarching philosophy, the UX Design Process is the actionable framework, and Mind Mapping is one of the first practical tools used to bridge the gap between user research and interface structure.

User-Centered Design (UCD): The Philosophy

UCD is not just a process; it is an internationally recognized mindset (ISO 9241-210) that places the end-user's needs, limitations, and environment at the center of every single decision. Instead of designing a digital product based on what the business wants to sell or what developers find easiest to build, UCD forces teams to design based on human behavior.

The philosophy relies on four key principles:

1. **Context of Use:** Identifying exactly who will use the product, what their primary goals are, and under what physical or emotional conditions they are using it.
2. **Requirements:** Defining the intersection where user goals meet business objectives.
3. **Design Solutions:** Moving from abstract concepts to concrete digital interfaces.
4. **Empirical Evaluation:** Validating those solutions against the original requirements by watching real people attempt to use the system.

The 5-Step UX Design Process

The practical application of UCD is typically broken down into a five-stage methodology (often popularized as Design Thinking). While it is taught linearly, in reality, teams constantly jump back and forth between these steps as they learn new information.

1. Understand (Learn About the User)

This step is about stepping outside your own perspective. As creators, we often assume we know exactly what our customers want. This step is about testing those guesses by actually talking to people and watching how they behave.

- **The Goal:** Find their hidden needs—the things users want but might not know how to explain clearly.
- **The Trekking Example:** You might guess a user booking an Annapurna trek cares most about the price. But when you talk to them, you find out they are actually really anxious. Their biggest fears are altitude sickness and not being physically fit enough. If you only design a simple "Book Now" page, you fail; you need to design a way to make them feel safe and prepared.

2. Define (Make Sense of the Mess)

You've collected a lot of notes from talking to people. This step forces you to summarize all that scattered information into one clear problem to solve. This is where you map out the user's feelings and actions step-by-step.

- **The Goal:** Create a shared focus for your team so everyone is solving the exact same problem.
- **The Trekking Example:** You summarize your research into one core problem: *"Users need to easily see how hard and safe a trek is before spending their money."* You map out their emotional journey—from the excitement of seeing a beautiful video to the fear they feel right before paying—finding the exact moment they usually leave the website.

3. Brainstorm (Explore Lots of Ideas)

Now that you know the exact problem, you come up with as many solutions as possible. You avoid using heavy design software here so you don't waste time trying to make things look perfect right away.

- **The Goal:** Focus on quantity over quality. You want obvious ideas, crazy ideas, and everything in between.
- **The Trekking Example:** How do we make users feel safe?
 - *Idea 1:* A direct WhatsApp chat button to talk to a guide.
 - *Idea 2:* A clean, grid-style layout that clearly shows daily oxygen levels.
 - *Idea 3:* A video slider of past hikers talking about how they handled the altitude.

You look at these to see what is actually possible to build, and you decide to go with the clean, grid-style layout.

4. Prototype (Build a Draft)

Brainstorming is about exploring options; prototyping is about narrowing down and building the best one. You take your winning idea and build a draft version. This goes from rough sketches to a screen that people can actually click on.

- **The Goal:** Build just enough of the design to test the idea without spending days or weeks making it perfect.
- **The Trekking Example:** You start with simple gray boxes to plan where things go. Once that makes sense, you make it look real. You use modern, clean styles—like soft colors and easy-to-read text—so the difficult mountain data looks beautiful and simple, rather than like a confusing science chart.

5. Test (The Reality Check)

A prototype is just a good guess; testing is the actual experiment. You put your clickable draft in front of real people and ask them to use it while you watch where they get stuck.

- **The Goal:** Find confusing parts before you spend time actually coding the website.
- **The Trekking Example:** You run a test where users try to book the Annapurna trip on their own. You track where they click and watch recordings of their screen. You notice users staring at your beautiful mountain data charts, but when they reach the payment screen, they get scared and close the window. The feedback shows that while they felt safe about the hike, they couldn't find your refund policy.

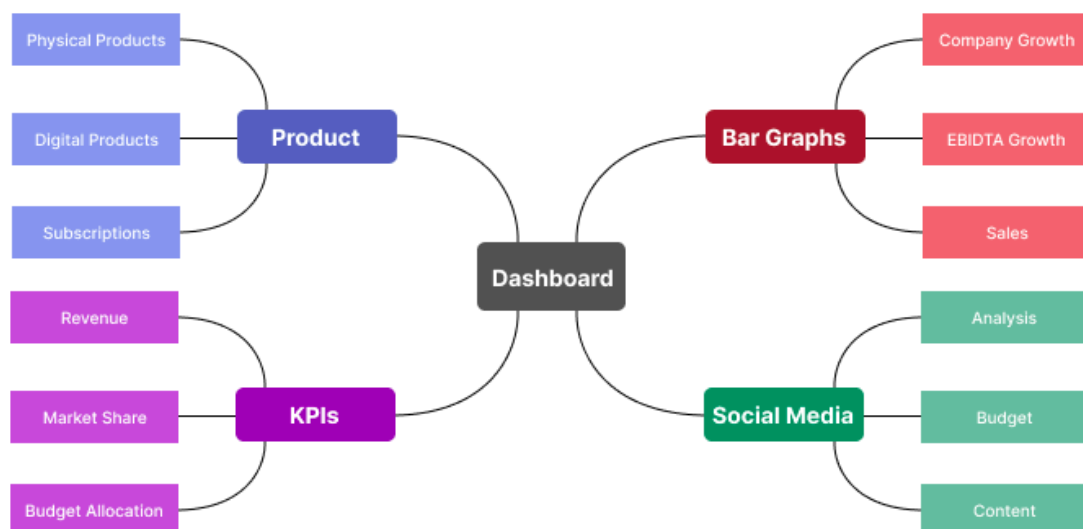
Key insight: The process is an infinite loop, not a straight line. Testing a prototype almost always sends the designer back to the Ideation or Definition phase to fix usability issues before the product is handed off to developers.

Mind Mapping: The Bridge to Ideation

When a designer finishes the "Define" phase, they are usually overwhelmed with data—dozens of user needs, feature requests, and business rules. A **Mind Map** is a visual diagram used during the "Ideate" phase to organize this chaos.

It starts with a single central concept (for example, a "User Dashboard") and branches outward into related categories, tasks, and sub-menus.

Here is an example of a mind map organizing the architecture of a complex dashboard before any wireframes are drawn:



Ux Information architecture mind map. Source: Geeksforgeeks

Why it matters: Mind mapping is the very first step toward creating Information Architecture (IA). It groups related concepts (e.g., placing "Promo Codes" and "Credit Cards" under a main "Checkout" branch). This dictates what information goes in a primary navigation bar versus what gets tucked away in a utility menu, ensuring users never get lost.

UX research: Conducting user research: Interviews, Surveys, Competitive analysis, creating user personas, user journey mapping

UX Research is broadly split into two distinct actions: **Gathering Data** (Interviews, Surveys, Competitive Analysis) and **Synthesizing Data** (Personas, Journey Mapping).

Part 1: Gathering Data (The Investigation)

Before you design a single screen, you must understand the problem space. Designers use three primary methods to gather data, each serving a completely different purpose.

1. User Interviews (Qualitative Research) Interviews are one-on-one conversations designed to uncover the “Why” behind user behavior. They are not meant for gathering statistics; they are for deep psychological insights.

- **The Technique:** Ask open-ended questions. Instead of asking, "Do you like booking tours online?" (which results in a yes/no), a UX researcher asks, "Tell me about the last time you tried to book a multi-day trip online. What frustrated you?"
- **The Goal:** Uncover hidden pain points, emotional triggers, and exact terminology the user uses (which directly informs Content Strategy later).

2. Surveys (Quantitative Research) Surveys are questionnaires sent to a large group of people to uncover the “What”. They validate the assumptions you generated during interviews.

- **The Technique:** Use closed-ended questions, Likert scales (1 to 5 ratings), and multiple choice.
- **The Goal:** Achieve statistical significance. If one person in an interview says they hate hidden fees, that is an anecdote. If a survey of 500 travelers reveals that 82% abandon checkout due to hidden fees, that is a data-backed UX priority.

3. Competitive Analysis You do not have to reinvent the wheel. Competitive analysis involves auditing direct rivals (other Himalayan trekking platforms) and indirect rivals (general travel platforms like Airbnb or Booking.com).

- **The Technique:** Perform a UX audit of their platforms. Identify common UI conventions (e.g., where do they place the filter bar?) so you don't break established user habits.

- **The Goal:** Find the gaps. If every competitor's website has a cluttered, text-heavy itinerary page, your competitive advantage becomes offering a clean, visually-driven bento-box layout.

Part 2: Synthesizing Data (Making it Actionable)

Once you have piles of interview transcripts and survey spreadsheets, you have to distill that data into tools the design team can actually use.

4. Creating User Personas A persona is a fictional character, but they are built entirely from real data. They serve as a proxy for your target audience, keeping the design team focused on the user rather than their own personal preferences.

A robust UX persona typically includes:

- **Demographics & Psychographics:** Age, tech-savviness, and lifestyle.
- **Primary Goals:** What are they specifically trying to achieve? (e.g., "Wants to book a safe, all-inclusive trek without managing daily logistics.")
- **Pain Points:** What stops them from achieving that goal? (e.g., "Overwhelmed by hidden costs and lack of visual proof of accommodations.")

5. User Journey Mapping A user journey map is a visual timeline that tracks the user's relationship with a product over time, from their initial discovery to their final action.

The map is usually divided into distinct phases (e.g., Discovery, Research, Booking, Post-Trip) and tracks three key metrics at each stage:

- **Touchpoints:** Where is the user interacting with the brand? (Instagram ad, Google search, Checkout screen).
- **Actions:** What is the user physically doing? (Comparing prices, reading reviews).
- **Emotions:** This is the most critical part. Is the user feeling excited, confused, or frustrated at this specific touchpoint?

Why Journey Mapping Matters: If the journey map shows that a user's emotional state drops from "Excited" to "Frustrated" specifically during the checkout phase, the UX team knows exactly where to focus their redesign efforts to prevent cart abandonment.

Ideation techniques: using Mood boards, Brainstorming and sketching

The primary rule of the ideation phase is **divergent thinking**: prioritizing the sheer quantity of ideas over the quality of any single idea.

1. Brainstorming: Breaking the Obvious

Brainstorming is about moving past the first, most obvious solution. When designing a platform for something as complex as a 14-day Everest Base Camp trek, the immediate assumption might be to just build a massive dropdown form. Brainstorming helps you discover better ways to solve that interaction.

Common UX Brainstorming Techniques:

- **"How Might We" (HMW) Questions:** Reframe pain points into opportunities. If research shows users abandon checkout because of hidden permit fees, you ask: *"How might we make government permit fees feel like a premium, transparent part of the experience rather than a hidden cost?"*
- **Crazy 8s:** A rapid-fire exercise where a designer takes a piece of paper, folds it into eight rectangles, and forces themselves to sketch eight distinct variations of a single screen (like a booking dashboard) in exactly eight minutes.
- **The Worst Possible Idea:** Sometimes, intentionally trying to design the absolute worst, most frustrating UI can highlight exactly what you need to avoid, sparking reverse-engineered solutions.

2. Mood Boards: Establishing the Visual Direction

Before you spend hours writing Tailwind CSS utility classes or building complex component variants, you need to ensure everyone (designers, developers, and clients) is aligned on the emotional feel of the product.

A mood board is a curated visual collage that dictates the aesthetic atmosphere before any actual UI layout is designed.

What goes into a UX/UI Mood Board?

- **Typography Pairings:** Selecting font families (like a clean sans-serif for Apple-style minimalism).
- **Color Palettes:** Defining the primary, secondary, and accent colors (e.g., using a crisp off-white background paired with a Himalayan sunset orange accent).
- **UI Textures & Patterns:** Snippets of specific design trends you want to utilize, such as examples of subtle glassmorphism (background blurs) or soft drop shadows.
- **Imagery Style:** Establishing the photographic direction. For a trekking platform, this might mean strictly using high-contrast, expansive drone shots rather than generic stock photos of backpackers.

3. Sketching: Low-Fidelity Layouts

Once the mood is established and ideas are brainstormed, sketching bridges the gap between abstract thought and digital architecture.

This is the physical act of drawing screen layouts on paper (or a digital whiteboard) using simple boxes, lines, and scribbles.

Why Paper Sketching is Critical:

- **Speed over Precision:** If you start directly in digital software, it is incredibly easy to get distracted by pixel-perfect alignment, padding, and hex codes. Sketching keeps you focused entirely on Information Architecture and user flow.
- **Exploring UI Patterns:** This is where you test out macro-structures. You might sketch one version of a homepage using a standard vertical list of tours, and another version using an interlocking bento-box grid to see which layout best highlights the photography.
- **Disposable Nature:** If a paper sketch doesn't work, you simply crumple it up. There is zero emotional attachment to the design, making it easier to pivot quickly before development begins.